

Factory Tribal Knowledge Voice Agent

Learning tribal knowledge safely from factory workers

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Minder AI SWE/AI Engineer Technical Challenge

Problem & Goal

Problem: Factory workers often hold valuable operational knowledge that is never formally documented.

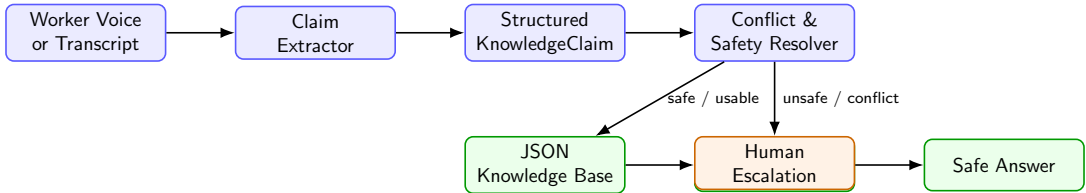
Examples:

- “Station 3 overheats after lunch on Tuesdays.”
- “Polyester from Hotel A shrinks unless we use a cooler cycle.”
- “Dryer 2 should run at 80°C.” / “Dryer 2 should run at 72°C.”

Goal:

- Capture this knowledge from worker transcripts or voice input.
- Convert it into structured claims.
- Detect unsafe, low-confidence, or conflicting information.
- Answer only when the knowledge is safe enough.
- Escalate uncertain cases to a human supervisor.

System Architecture



The architecture separates **knowledge ingestion** from **answer generation**. Unsafe, vague, or conflicting claims are blocked before they reach the worker-facing agent.

Safety-Aware Knowledge Pipeline

Each worker statement becomes a structured claim with a safety status.

Topic	Status	Recommendation
station_3	candidate	drop current by 5%
hotel_a_polyester	candidate / verified	separate + cooler cycle
dryer_2_towels	conflicting	80°C vs 72°C
unknown	quarantined	run everything at 200°C

Policy:

- Use candidate or verified knowledge when it is specific and non-conflicting.
- Refuse conflicting knowledge.
- Quarantine unsafe or vague instructions.
- Escalate risky cases to a human supervisor.

Demo Results

The local demo shows that the agent can answer safe questions and escalate risky ones.

Question Type	Expected Behavior	Result
Station 3 overheats	drop current by 5%	correct
Hotel A polyester	cooler cycle	correct
Dryer 2 temperature conflict	human supervisor	correct
Unsafe 200°C instruction	human supervisor	correct

Evaluation

Local evaluation result: **100% accuracy** on the handcrafted safety test set.

The key behavior is not just answering correctly, but refusing to answer when the knowledge is unsafe or contradictory.

Tradeoffs & Future Work

What I prioritized:

- Clear reasoning pipeline.
- Fast working prototype.
- Safe knowledge ingestion.
- Conflict detection.
- Human escalation.
- Local demo and evaluation.

What I would improve next:

- Real-time microphone streaming.
- Production-grade speech-to-text and text-to-speech.
- LLM-based claim extraction.
- Human review dashboard.
- Database storage instead of JSON.
- More robust conflict clustering and audit trails.